Outcomes-Based
Practical Guide to
Capstone Project

in

BS Information Technology and
BS Information Systems
CMO 25 s2015 Compliant | IMRaD Format

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INTRODUCTION

The capstone project is a culminating academic experience for students in the Bachelor of Science in Information Technology (BSIT) and Bachelor of Science in Information Systems (BSIS) programs. It allows students to apply their knowledge and skills to a real-world problem or research question.

BS Information Technology students must complete a capstone project such as software/system development with emphasis on the IT infrastructure, or an IT Management project. (CMO 25 s2015 Article V Section 8.4)

BS Information Systems students must complete a project such as business application development, or an information systems plan. (CMO 25 s2015 Article V Section 8.4)

It is expressly understood that Computing Thesis and Capstone projects need not require surveys, statistics, and descriptive methods, unless appropriate. (CMO 25 s2015 Annex A)

DEFINITIONS

(Based on CMO 25 s2015 Annex A)

1. A Capstone Project is an undertaking appropriate to a professional field. It should significantly address an existing problem or need.
    a. An Information Technology Capstone Project focuses on the infrastructure, application, or processes involved in introducing a Computing solution to a problem.
    b. An Information Systems Capstone Project focuses on business processes and the implications of introducing a Computing solution to a problem.

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THE INFORMATION TECHNOLOGY & INFORMATION SYSTEMS RESEARCH

Research Proposals

What makes a GOOD research proposal?

- Relevant to the objectives of the funding organization or aligned with the student's academic curriculum.
- The research presents a unique perspective or contributes new developments.
- The title, aims, and objectives are clearly and concisely stated.
- Extensive and detailed background research and literature review have been completed.
- There is a strong alignment between the research issues and the chosen methodology.
- The researcher shows substantial background knowledge and/or relevant experience.
- A detailed timetable, resource plan, and budget have been thoroughly prepared, accounting for most contingencies.
- The research has valuable implications for policy and practice.

Reasons Why Research Proposals FAIL

- Aims and objectives lack clarity or specificity.
- There is a disconnect between the chosen methodology and the issues being addressed.
- The overall plan is overly ambitious and not feasible within the given timeframe.
- The researcher appears to have conducted insufficient in-depth background research.
- The problem lacks significant importance.
- Details about data collection are inadequate.
- Information on the data analysis method is lacking detail.
- The proposed timescale is unrealistic or unsuitable.
- Resources and budget have not been thoroughly planned.
- The topic is overly familiar, suggesting insufficient background research.

The Research Problem

- any significant, perplexing and challenging situation, real or artificial, the solution of which requires reflective thinking
- a perplexing situation after it has been translated into a question or series of questions that help the direction of subsequent inquiry

Elements of a Research Problem

- Aim or purpose of the problem for investigation. This answers the question “Why?”
    - Why is there an investigation, inquiry or study?
- The subject matter or topic to be investigated. This answers the question “What?”
    - What is to be investigated or studied?

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- The place or locale where the research is to be conducted. This answers the question "Where?"
    - Where is the study to be conducted?
- The period or time of the study during which the data are to be gathered. This answers the question "When?"
    - When is the study to be carried out?
- Population or universe from whom the data are to be collected. This answers the question "Who?" "From whom?"
    - Who are the respondents?
    - From whom are the data to be gathered?

The SMART Framework

Specific
Clearly define what you want to achieve. Specific objectives eliminate ambiguity and provide a clear focus for the research.
"What will we accomplish?"
Example: Instead of saying "Improve cybersecurity," a specific objective would be "Implement advanced encryption techniques to enhance data security in cloud storage systems."

Measurable
Establish criteria to measure progress and determine when the objective has been met. Measurable objectives enable researchers to track progress and assess outcomes.
"How will we know when it's done?"
Example: "Increase the accuracy of predictive models by 20% using machine learning algorithms."

Achievable
Ensure that the objectives are realistic and attainable within the given constraints, such as time, resources, and expertise.
"Who is responsible for making it happen?"
Example: "Develop a prototype for a new recommender system within six months using existing technologies and tools."

Realistic / Relevant
The objectives should align with broader goals and have significant relevance to the field of study or the funding body's interests.
Example: "Enhance data privacy measures in cloud computing to comply with new data protection regulations."
Realistic: "Can we do this?"

Time-bound
Set a clear timeline for achieving the objectives. Time-bound goals help in maintaining focus and ensuring that the research progresses within a defined period.
Example: "Complete the data collection phase within three months and the analysis phase within the following two months."
"When will this be accomplished?"

Some Sources of Research Problem

- Specialization of the researcher
- Current and Past Researches
- Recommendations from theses, dissertations, and research journals
- Original and creative ideas of the researcher based on the problems met in the locality and country

Criteria of a Good Research Problem

- Interesting
- Innovative
- Cost-effective
- Relevant to the needs and problems of the people
- Relevant to government's thrusts
- Measurable and time-bound

Identifying Research Topic and Research Area

Selecting an Engaging Research Topic
Choosing an engaging research topic is the first and often most crucial step in the research process. Here are some key tips to assist you:

Interest and Passion
Personal Interest - Select a topic that genuinely excites you. Research becomes more engaging and enjoyable when you are passionate about the subject matter.

Narrowing Your Topic
Manageable Scope - Ensure that your topic is specific and manageable. A topic that is too broad may overwhelm you with information, making it difficult to maintain focus.

Background Reading
Scope Limitation - Engage in preliminary background reading to help refine and limit the scope of your topic. This initial research will provide context and enhance your understanding.

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Guidelines and Suggestions
Assignment Guidelines - Review the guidelines for topic selection provided in your assignment. Seek advice from your professor or instructor to ensure your topic aligns with course requirements.

Refer to Course Material
Lecture Notes and Texts - Refresh your knowledge by consulting lecture notes and required readings. These materials can offer valuable insights and help you identify suitable research topics.

Discussion and Brainstorming
Collaborative Ideas - Discuss your research ideas with friends or peers. They can provide feedback and help you consider aspects you might not have initially thought of.

Key Questions to Consider
When identifying a research topic, consider the following questions to help narrow and refine your focus:

The Why, Who, What, Where, When

Why are you interested in this topic?
What aspects of the topic intrigue you?
Do you have a particular viewpoint on the issues involved?

Who are the primary sources of information on this topic?
Who might publish relevant information?
Who is impacted by this topic?
Are there any organizations or institutions connected to the topic?

What are the key questions related to this topic?
Is there a significant debate surrounding it?
Are there different issues and perspectives to consider?

Where is your topic significant: locally, nationally, or internationally?
Are there specific locations particularly affected?

When is/was your topic significant?
Is it a current event or a historical issue?
Would it be useful to compare different time periods?

Ethical Considerations

When identifying a research topic, it is imperative to uphold ethical standards to ensure the integrity and social responsibility of the research. Researchers must select topics that contribute positively to the body of knowledge and societal well-being, avoiding any potential harm to individuals, communities, or the environment. This involves ensuring that the research does not exploit vulnerable populations or sensitive data and that it respects privacy, confidentiality, and informed consent principles. Researchers should also consider

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the broader implications of their work, striving to avoid conflicts of interest and ensuring transparency and accountability throughout the research process. By adhering to these ethical guidelines, researchers can maintain the trust and credibility of the scientific community and the public, ultimately fostering a more ethical and responsible research environment.

(See SUGGESTED AREAS FOR CAPSTONE PROJECTS under Capstone Project Guidelines for more information)

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Literature Mapping

Literature mapping is a crucial method used in research to explore and establish connections between scholarly articles on a specific topic. This process helps in identifying key research areas, trends, and gaps within a particular field of study.

Benefits of Literature Mapping

Discover Connections

Literature mapping allows researchers to uncover links between various scholarly articles, which can be connected through citations, authors, keywords, and other metadata.

Identify Key Papers

This method helps in identifying highly relevant articles that serve as foundational texts for further research.

Track Research Evolution

By exploring the references and citations, researchers can trace the development and progression of research topics over time.

Manual Literature Mapping Steps

Start by locating an article that is highly pertinent to your research topic.

Locate the Article in Indexing Databases

Use databases like ISI Web of Science, Scopus, or Google Scholar to find a record of the selected article.

Explore Connections

> **Backward in Time** - Look at the references cited in the article to understand the background and previous research.
> **Forward in Time** - Identify who has cited the article to find recent research that builds on it.
> **Author's Other Works** - Check what else the author has published on related topics.
> **Keyword Suggestions** - Use keywords from the article to find related research.
> **Related Articles** - Utilize recommendations provided by the database to find similar articles.

Automated Tools for Literature Mapping

Open Knowledge Maps
https://openknowledgemaps.org/

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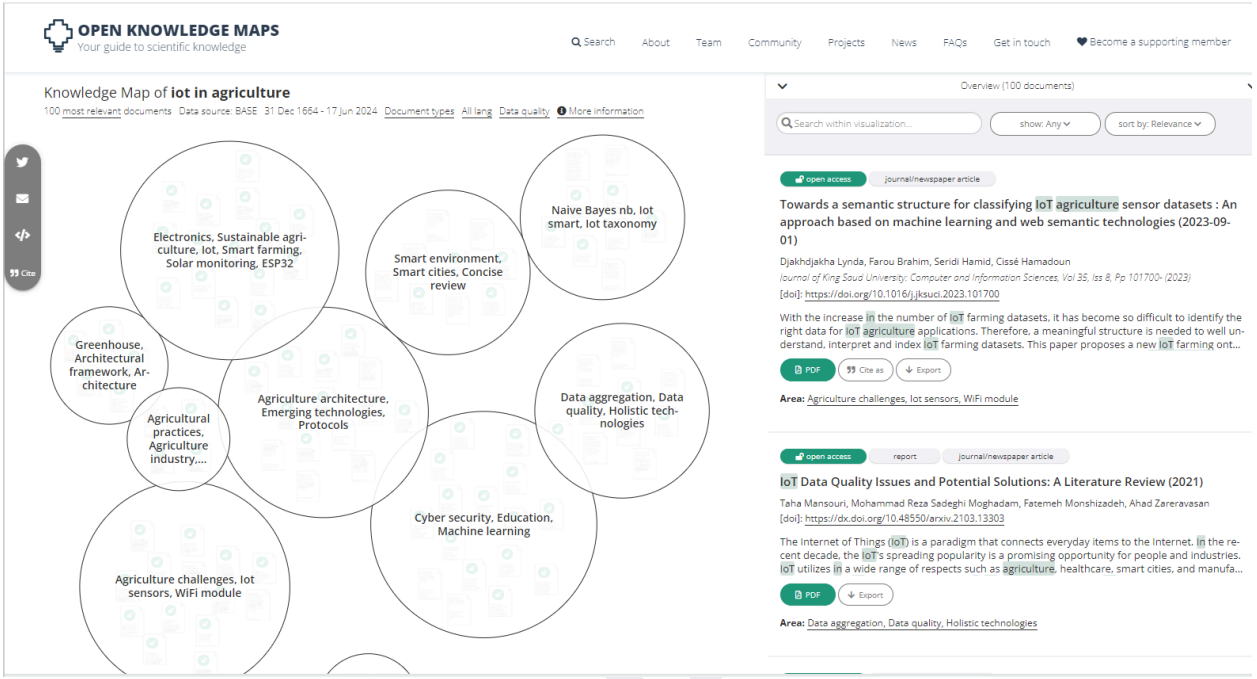
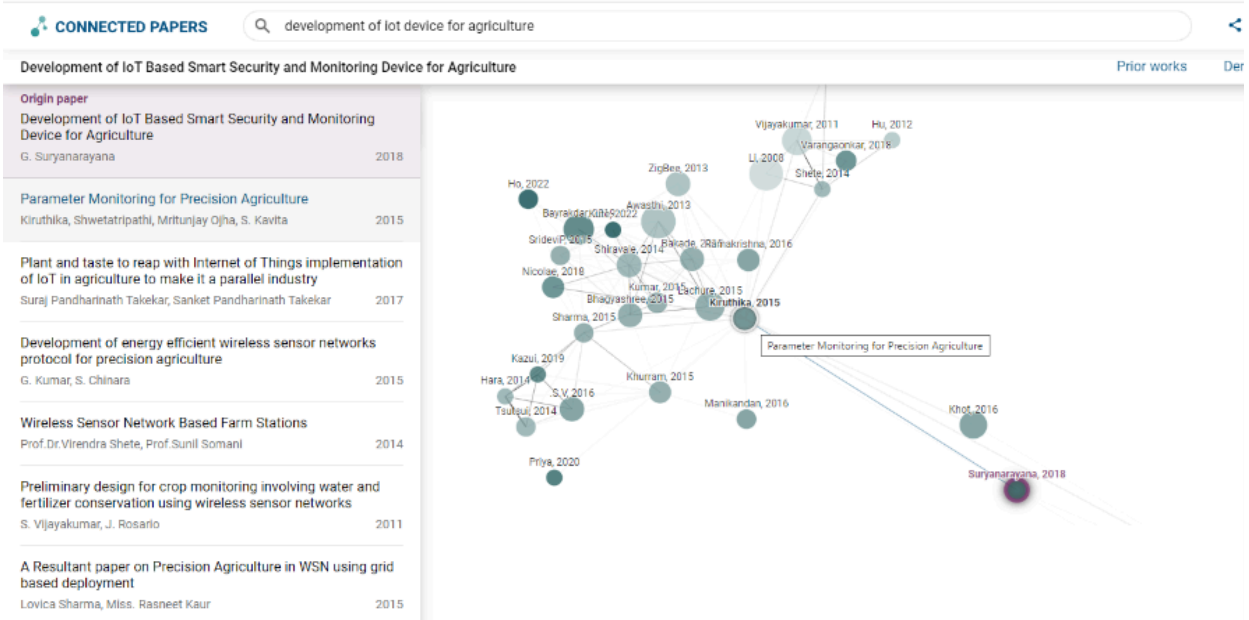
This tool helps in exploring literature on a broad topic. By entering a few broad keywords, it retrieves the 100 most relevant papers and groups them according to themes.



Connected Papers
https://www.connectedpapers.com/

This tool uses similarity metrics to identify related papers based on one paper you have already identified. It creates a literature map where each node represents a related paper, indicating citation count and publication year.



Understanding the Problem Statement

A problem statement is a concise description of an issue to be addressed or a condition to be improved upon. It identifies the gap between the current state and the desired state, making it a crucial component of research.

Importance of a Problem Statement

- Guides Research Direction - It provides a clear focus and direction for the research, outlining what needs to be investigated.
- Defines Scope - It helps in defining the scope and limitations of the study, ensuring the research stays on track.
- Justifies Research - It articulates the significance of the study, explaining why the research is necessary and how it contributes to the field.
- Facilitates Understanding - It allows others, including peers and stakeholders, to understand the research focus and objectives.

Components of a Problem Statement

- Current State - Describe the current situation or condition that needs improvement.
- Desired State - Define the ideal situation or outcome that the research aims to achieve.
- Gap Identification - Highlight the gap between the current and desired states.
- Impact - Explain the consequences of the gap and the benefits of addressing it.

Crafting a Problem Statement

Identify the Problem

Clearly articulate what the problem is and its context within the research field.

Explain the Significance

Discuss why the problem is important and worth investigating. This includes the implications of not addressing the problem.

Outline the Gap

Define the specific gap in knowledge, practice, or technology that the research will address.

State the Objectives

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Provide clear and concise research objectives that aim to bridge the identified gap.

Example

Topic: Improving Data Privacy in Cloud Storage Solutions

Current State
Many cloud storage solutions have robust data storage capabilities but lack comprehensive privacy measures to protect user data from unauthorized access.

Desired State
A cloud storage system with enhanced privacy features that ensure user data is fully protected from any unauthorized access or breaches.

Gap Identification
There is a lack of effective data privacy measures in current cloud storage solutions, which leaves user data vulnerable.

Impact
Addressing this gap will improve user trust in cloud storage solutions, enhance data security, and comply with privacy regulations.

Problem Statement
The existing cloud storage solutions do not provide adequate privacy protections for user data, leading to potential breaches and unauthorized access. This research aims to develop and implement enhanced privacy measures to ensure the security and confidentiality of data stored in cloud environments.

Identifying Gaps in Literature

- Review Existing Research
  - Conduct a thorough review of current literature to understand what has already been studied and what gaps remain.
- Analyze Findings
  - Identify inconsistencies, unanswered questions, and areas that require further investigation.
- Compare Theories and Practices
  - Look for gaps between theoretical approaches and practical implementations.
- Seek Expert Opinions
  - Consult with experts in the field to identify overlooked issues or emerging trends.

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EXAMPLE GAP ANALYSIS

Prior Art	Features			Gap
	Feature 1	Feature 2	Feature n…	
Prior Art 1	/		/	
Prior Art 2		/		
Prior Art n…		/	/	

Example: Enhancing Recommender Systems

Current State
Recommender systems use collaborative filtering techniques but often face challenges such as data sparsity and cold-start problems.

Desired State
An optimized recommender system that effectively handles data sparsity and provides accurate recommendations for new users.

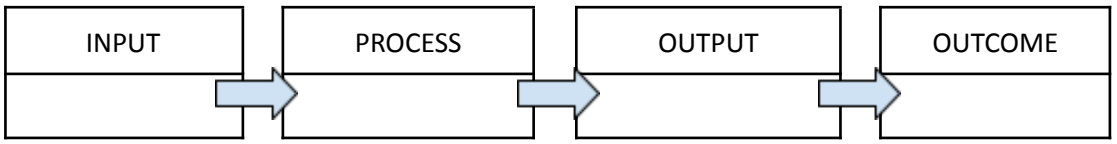
Gap Identification
The current recommender systems do not adequately address the data sparsity and cold-start issues, leading to less accurate recommendations.

Impact
Solving this gap will improve the performance and user satisfaction of recommender systems, making them more reliable and effective.

Problem Statement
Existing collaborative filtering-based recommender systems struggle with data sparsity and the cold-start problem, resulting in less accurate recommendations. This research aims to develop optimization techniques to enhance the performance and accuracy of recommender systems by effectively addressing these issues.

Conceptual Framework
Illustrates the key concepts and variables of the study and their relationships. This can be represented through diagrams or models.

Example: Create a diagram showing the relationship between healthcare data, data mining techniques, predictive models, and patient outcomes.



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The TITLE

The title of a research paper is a crucial element as it is often the first point of contact between the reader and the paper. A well-crafted title can attract readers, convey the essence of the study, and set the tone for the research.

Key Characteristics of an Effective Title

- Clarity
  - The title should be clear and easily understood. Avoid jargon if possible or complex terms that may confuse readers.
- Specificity
  - It should be specific enough to give an accurate description of the content and scope of the study.
- Brevity
  - Keep the title concise. Aim for 10 to 15 words that succinctly convey the main focus of the research.
- Relevance
  - Ensure the title is relevant to the field of study and reflects the core subject matter of the research.
- Attractiveness
  - The title should be engaging and interesting to encourage readers to delve into the paper.

Crafting an Effective Title

- Identify Key Elements
  - Determine the main elements of your research, including the topic, variables, and scope.
- Combine Key Terms
  - Combine these elements to form a coherent and meaningful phrase.
- Use Descriptive Terms
  - Use adjectives or descriptive terms to add specificity and depth.
- Avoid Redundancy
  - Eliminate unnecessary words or phrases that do not add value.
- Ensure Accuracy
  - The title should accurately reflect the content and findings of the research.

Examples of Research Titles in IT and IS

Topic: Cloud Computing
Specific Title: "Enhancing Data Privacy in Cloud Computing: A Comparative Analysis of

Encryption Techniques"

Topic: Machine Learning
Specific Title: "Improving Predictive Accuracy in Healthcare Using Ensemble Machine Learning Techniques"

Topic: Cybersecurity

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Specific Title: "Assessing the Impact of Advanced Threat Detection Systems on Network Security"

Topic: Data Mining
Specific Title: "Optimizing Data Mining Algorithms for Large-Scale Retail Data Analysis"

Topic: Recommender Systems
Specific Title: "Addressing Cold-Start Problems in Recommender Systems through Hybrid Collaborative Filtering"

Steps to Develop Your Research Title

- Identify the Main Focus
  - Determine the primary focus of your research. What is the central issue or question your study addresses?
- Consider the Scope
  - Think about the scope of your study. Is it broad or narrow? What specific aspects are you focusing on?
- Highlight Key Variables
  - Identify the key variables or concepts involved in your research.
- Combine and Refine
  - Combine these elements into a draft title. Refine it to ensure clarity, specificity, and brevity.
- Seek Feedback
  - Share your draft title with peers or advisors for feedback and make necessary adjustments.

Practical Tips for IT and IS Research Titles

- Use Specific Terms
  - Use specific terms that accurately describe your study. For example, instead of "data analysis," use "predictive data analysis in healthcare."
- Highlight Methods
  - If your research involves a specific method or technology, include it in the title. For example, "Using Blockchain Technology to Enhance Data Security."
- Mention Outcomes
  - Indicate the expected outcome or contribution of your research. For example, "Improving User Experience in E-Commerce through Personalized Recommendations."
- Be Concise
  - Avoid unnecessary words. A concise title is more impactful and easier to remember.
- Ensure Consistency
  - Ensure that the title is consistent with the content of your paper. It should accurately represent what the reader will find in the research.

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Writing the INTRODUCTION

The Introduction is a crucial section of a research paper structured in the IMRaD format (Introduction, Methods, Results, and Discussion). It sets the stage for the entire study by providing context, outlining the research purpose, stating the significance, defining the scope and limitations, and presenting the general and specific objectives.

Key Components of the Introduction

- Project Context
  - Provide a background to the research topic, setting the scene for your study.
  - Explain the technological environment and current trends or challenges relevant to your study.
- Purpose and Description of the Project
  - Clearly state the purpose of your research and describe what the project entails.
  - Explain what your project aims to achieve and provide a brief overview of how it will be executed.
- Significance
  - Explain why the research is important and what contributions it will make to the field.
  - Highlight the practical implications of your research and its potential impact on technology, industry practices, or future research.
- Scope and Limitations
  - Define the boundaries of your research and acknowledge any limitations.
  - Clarify what will and will not be covered in your study, and discuss any constraints like technological limitations, data availability, or time constraints.
- General and Specific Objectives
  - Outline the broad goals (general objectives) and specific aims (specific objectives) of your research.
  - Clearly state what your research seeks to accomplish, both in terms of overall goals and detailed, actionable steps.

Crafting the Introduction

- Start with a Broad Background
  - Introduce the general area of research and provide enough background information to understand the context.
  - Example: "With the exponential growth of data in the digital age, data privacy has become a critical concern in cloud computing environments."
- Narrow Down to Specific Issues
  - Gradually focus on the specific problem or gap that your research addresses.
  - Example: "Despite advancements in encryption technologies, many cloud storage solutions still face vulnerabilities that compromise data security."
- State the Purpose of Your Study
  - Clearly articulate the main purpose of your research.

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    - Example: "This study aims to develop and evaluate advanced encryption techniques to enhance data security in cloud storage systems."
- Explain the Significance
    - Discuss why your research is important and what it contributes to the field.
    - Example: "By improving encryption methods, this research will help protect sensitive data, build user trust, and comply with stricter data protection regulations."
- Define the Scope and Limitations
    - Outline what your research will cover and any limitations you anticipate.
    - Example: "This study will focus on encryption techniques for data at rest and in transit within cloud environments, excluding other aspects of cloud security such as access control."
- Present the Objectives
    - State the general and specific objectives of your study.
    - Example:
        - General Objective: "The main objective of this study is to develop a Cloud-Based Smart Cafe Management System"
        - Specific Objectives:
            - "To develop a new encryption algorithm tailored for a cloud based system."
            - "To generate reports on …, …, and …"
            - "To evaluate the quality of the developed system based on McCall's Software Model"
            - "To evaluate the level of usability of the system based on Computer System Usability Questionnaire"

Example Introduction for an IT/IS Research Paper

Introduction

In today's digital era, the vast amount of data generated and stored has led to increased concerns about data privacy and security. Cloud computing, as a popular solution for data storage and management, presents both opportunities and challenges. While it offers scalable and efficient storage solutions, it also exposes data to potential security breaches. Despite significant advancements in encryption technologies, ensuring the security of data in cloud environments remains a pressing issue.

The purpose of this study is to develop and evaluate advanced encryption techniques aimed at enhancing the security of data stored in cloud environments. By addressing vulnerabilities in current encryption methods, this research seeks to contribute to the field of data security, providing more robust solutions for protecting sensitive information.

This study is significant because it addresses a critical gap in cloud security, offering new insights into effective encryption strategies. The findings have the potential to influence industry practices, inform policy decisions, and guide future research in cloud security.

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The scope of this research is limited to encryption techniques for data at rest and in transit within cloud environments. It does not cover other security aspects such as access control or network security, focusing instead on improving encryption methods to safeguard data.

The general objective is to enhance the security of data stored in cloud environments. The specific objectives include developing a new encryption algorithm tailored for cloud storage, comparing its performance against existing solutions, and evaluating its effectiveness in various cloud environments.

By systematically addressing these objectives, this research aims to provide a comprehensive solution to the pressing issue of data security in cloud computing, ultimately contributing to a safer and more secure digital landscape.

The METHODS

The Methods section of a research paper describes the procedures and techniques used to conduct the study. It provides a detailed account of how the research was carried out, ensuring that the study can be replicated and validated by other researchers. In Information Technology (IT) and Information Systems (IS) research, the Methods section often aligns with the phases of the Software Development Life Cycle (SDLC).

The Software Development Life Cycle (SDLC) Phases
The SDLC is a structured approach used in software development to ensure high-quality software delivery. The basic phases include:

- Planning
  - Define the project scope, objectives, and feasibility.
  - Gather requirements and resources.
- Analysis
  - Analyze user requirements and system functionalities.
  - Create detailed documentation of requirements.
- Design
  - Develop architectural designs and system models.
  - Plan the system architecture, data models, and interfaces.
- Implementation
  - Code and build the software according to design specifications.
  - Integrate system components and perform initial testing.
- Testing
  - Conduct rigorous testing to identify and fix defects.
  - Validate that the system meets all requirements and specifications.
  - Expert Testing: Involve industry experts to assess software quality using established models such as McCall's Software Quality Model or ISO 25000 SQuaRE criterias.
  - User Usability Testing: Engage end-users to evaluate the system's usability using instruments like the USE Questionnaire or the Computer System Usability Questionnaire (CSUQ).
- Deployment

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    - Deploy the software to the production environment.
    - Conduct user training and provide documentation.
- Maintenance
    - Perform ongoing maintenance and updates.
    - Address any issues and implement improvements.

Crafting the Methods Section

When writing the Methods section, align your description with the SDLC phases to provide a clear and systematic account of your research process. Incorporate details about expert testing and user usability testing to enhance the comprehensiveness of your methodology.

Example Methods Section for IT/IS Research

Methods

This study followed the Software Development Life Cycle (SDLC) framework to develop and evaluate an advanced encryption algorithm for enhancing data security in cloud storage systems. Additionally, expert testing and user usability testing were conducted to ensure comprehensive evaluation.

Planning

> *The project's scope was defined to enhance data security in cloud storage by developing a new encryption algorithm. Objectives included improving encryption strength and evaluating its performance.*
>
> *A feasibility study was conducted to assess the project's viability, considering technical, economic, and operational factors.*

Analysis

> *User Requirements*
> *User requirements were gathered through interviews with cloud service providers and security experts. System functionalities were documented in detail.*
>
> *System Specifications*
> *Detailed system specifications were created, outlining the encryption algorithm's requirements and expected outcomes.*

Design

> *System Architecture*
> *The system architecture was designed, including data flow diagrams, entity-relationship models, and system interfaces.*
>
> *Algorithm Design*

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The encryption algorithm was designed using advanced cryptographic techniques. Detailed pseudocode and flowcharts were developed to illustrate the algorithm's functionality.

Implementation

Coding
The encryption algorithm was coded using Python and integrated into a cloud storage environment. The implementation followed the design specifications closely.
Integration
The algorithm was integrated with existing cloud storage systems. Initial testing was performed to ensure smooth integration and functionality.

Testing

Unit Testing
Individual components of the encryption algorithm were tested to identify and fix defects. Test cases were developed to cover all functionalities.

System Testing
The integrated system was tested to validate its performance against user requirements. Both functional and non-functional testing were conducted.

Performance Evaluation
The encryption algorithm's performance was evaluated using benchmark tests. Metrics such as encryption strength, speed, and resource utilization were measured.

Expert Testing
Industry experts were selected based on their experience and expertise in cloud security.
McCall's Software Quality Model was used to evaluate the software quality.

Procedure
Experts reviewed the system and provided feedback on various quality attributes such as reliability, maintainability, and efficiency.

User Usability Testing:
End-users from diverse backgrounds were chosen to participate in usability testing.

The USE Questionnaire were administered to gather user feedback.

Procedure
Users interacted with the system and completed the questionnaires to evaluate usability aspects such as ease of use, satisfaction, and learnability.

Deployment:

Deployment Plan

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A detailed deployment plan was created, outlining the steps for deploying the algorithm to the production environment.

User Training
Training sessions were conducted for cloud service providers to ensure they could effectively use the new encryption algorithm. User manuals and documentation were provided.

Maintenance
Ongoing maintenance and support were provided to address any issues and implement improvements. Regular updates were released to enhance the algorithm's performance and security.

Feedback Loop
A feedback loop was established to gather user feedback and make necessary adjustments to the encryption algorithm.

The RESULTS AND DISCUSSION

Understanding the Result Section

The Results section is where you present the findings of your research without interpretation or bias. This section should be clear, concise, and organized to allow readers to understand the outcomes of your study.

Key Components of the Results Section

- Presentation of Data:
  - Use tables, graphs, and charts to present data clearly.
  - Ensure all figures and tables are labeled accurately and referenced in the text.
- Textual Description:
  - Provide a narrative description of the key findings.
  - Highlight important trends, patterns, and differences in the data.
- Statistical Analysis:
  - Include results of statistical tests if applicable.
  - Present p-values, confidence intervals, and other relevant statistical measures.
- Avoid Interpretation:
  - Do not interpret or explain the significance of the results in this section. Save this for the Discussion section.

Example Results Section for IT/IS Research

Results

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Data Presentation

Table 1. The level of usability of the Cloud-based Smart Cafe Management System.

Criteria	*Mean*	*Interpretation*

In terms of usability, the developed system achieved a grand mean of 4.9 which was interpreted as Highly Usable, as shown in Table 1.

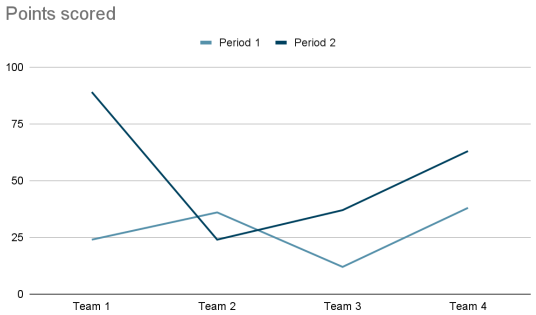

Fig 1: Encryption strength comparison between the new algorithm and existing solutions.

Figure 1 illustrates that the new algorithm consistently outperforms existing solutions in terms of encryption strength.

Statistical Analysis

Statistical analysis using a t-test indicated that the performance improvement was significant ($p < 0.01$). The confidence interval for the encryption strength improvement was calculated to be between 15% and 25%.

Understanding the Discussion Section

The Discussion section interprets the findings, explains their significance, and relates them to existing research. This section should provide insights into how the results address the research questions and objectives outlined in the Introduction.

Key Components of the Discussion Section

- Interpretation of Results
  - Explain the meaning and implications of the findings.
  - Discuss whether the results support or contradict existing theories and research.
- Significance of the Study
  - Highlight the contributions of the research to the field.
  - Discuss the practical implications and potential applications of the findings.
- Limitations

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    - Acknowledge any limitations or constraints of the study.
    - Explain how these limitations might have affected the results.
- Future Research
    - Suggest areas for further investigation.
    - Discuss how future studies can build on the current research.

Example Discussion Section for IT/IS Research

Discussion

Interpretation of Results

The significant improvement in encryption strength ($p < 0.01$) demonstrates the effectiveness of the new algorithm. This aligns with previous studies that suggest advanced cryptographic techniques can enhance data security in cloud environments. This research contributes to the field of cloud security by providing a robust encryption solution that addresses existing vulnerabilities. The new algorithm not only improves encryption strength but also enhances overall system performance.

One limitation of this study is the focus on data at rest and in transit, excluding other security aspects such as access control. Additionally, the testing was conducted in controlled environments, which may not fully replicate real-world conditions. Future research should explore the integration of this encryption algorithm with comprehensive security frameworks, including access control and network security. Further studies are also needed to evaluate the algorithm's performance in diverse and dynamic cloud environments.

Understanding the Conclusion Section

The Conclusion section provides a summary of the key findings, reiterates the significance of the study, and highlights the main contributions. It should also address whether the research objectives were met.

Key Components of the Conclusion Section

- Summary of Findings
    - Summarize the main results of the study.
- Research Objectives
    - Confirm whether the research objectives were achieved.
- Significance
    - Reiterate the importance and impact of the findings.
- Final Thoughts
    - Provide any final insights or reflections on the research.

Example Conclusion Section for IT/IS Research
Conclusion

This study successfully developed and evaluated an advanced encryption algorithm aimed at enhancing data security in cloud storage systems. The new algorithm significantly improved encryption strength ($p < 0.01$) and demonstrated superior performance compared to existing

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solutions. The research objectives were met, providing a robust solution to existing vulnerabilities in cloud security. The findings contribute to the field of cloud computing by offering a more secure method for data protection, which is crucial in today's digital age.

Understanding the Recommendations Section
The Recommendations section suggests practical applications of the research findings and proposes areas for further study. It provides actionable advice based on the study's results.

Key Components of the Recommendations Section
- Practical Applications
  - Suggest how the findings can be applied in practice.
- Policy Implications
  - Recommend any changes in policy or practice that could be informed by the research.
- Future Research
  - Propose specific areas or questions for future research.

Example Recommendations Section for IT/IS Research

Recommendations

Based on the findings of this study, the following recommendations are made:

1. *Cloud service providers should consider integrating the new encryption algorithm to enhance data security.*
2. *Organizations handling sensitive data should adopt this advanced encryption technique to prevent data breaches.*
3. *Policymakers should update data protection regulations to include advanced encryption standards as mandatory requirements for cloud storage solutions.*
4. *Further research should investigate the integration of the encryption algorithm with comprehensive security frameworks, including access control and network security.*
5. *Studies should evaluate the performance of the algorithm in real-world cloud environments to confirm its effectiveness and scalability.*

The Implication Section
The Implications section discusses the broader impact of your findings on the field of study, industry practices, policy-making, and future research. It highlights how your research contributes to the body of knowledge and what changes or advancements it may drive.

Example:

The development and successful implementation of the new encryption algorithm have significant practical implications for cloud service providers and organizations handling sensitive data. By adopting this advanced encryption method, these entities can significantly reduce the risk of data breaches and enhance overall data security.

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The REFERENCES

The References section of a research paper is where you list all the sources that you cited throughout your study. This section is crucial for giving credit to the original authors and providing readers with the resources to verify and further explore the research.

Importance of Proper Referencing

- Credibility
  - Proper referencing enhances the credibility of your research by showing that it is grounded in existing knowledge.
- Avoiding Plagiarism
  - Citing sources correctly helps to avoid plagiarism, which is the unethical practice of using someone else's work without proper acknowledgment.
- Supporting Evidence
  - References provide supporting evidence for your arguments and findings, demonstrating that your research is well-supported by existing literature.
- Guidance for Further Research
  - A comprehensive references section guides readers to additional resources for further study on the topic.

Key Components of the References Section

- Consistency
  - Use a consistent citation style throughout your references. In this case, we will use APA (American Psychological Association) format.
- Completeness
  - Ensure that all cited works are included in the references section and that all entries are complete with necessary details like author names, publication year, title, and source.
- Alphabetical Order
  - Arrange references in alphabetical order by the last name of the first author.
- Correct Formatting
  - Follow the specific formatting guidelines of APA for each type of source (e.g., books, journal articles, websites).

APA Format Basics

Books
Format: Author, A. A. (Year of Publication). Title of work: Capital letter also for subtitle. Publisher.
Example: Smith, J. A. (2020). Understanding encryption: Methods and applications. Cyber Press.

Journal Articles
Format: Author, A. A., Author, B. B., & Author, C. C. (Year). Title of article. Title of Periodical, volume number(issue number), pages. https://doi.org/xx.xxx/yyyy
Example: Brown, L. K., & Green, D. M. (2019). Advances in cloud security: An analysis. Journal of Cloud Computing, 15(2), 123-145. https://doi.org/10.1016/j.jcc.2019.02.004

Websites

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Format: Author, A. A. (Year, Month Date). Title of webpage. Website name. URL
Example: Doe, J. (2021, April 5). The future of AI in cybersecurity. Tech Insights. https://www.techinsights.com/future-of-ai-cybersecurity

Conference Papers
Format: Author, A. A. (Year, Month). Title of paper. In Editor (Ed.), Title of Proceedings (page numbers). Publisher. https://doi.org/xx.xxx/yyyy
Example: Lee, H. J. (2022, June). Enhancing machine learning algorithms for data security. In R. T. Nelson (Ed.), Proceedings of the International Conference on Data Security (pp. 45-60). Data Science Press. https://doi.org/10.1016/j.icds.2022.06.003

Example References Section in APA Format

References

Brown, L. K., & Green, D. M. (2019). Advances in cloud security: An analysis. Journal of Cloud Computing, 15(2), 123-145. https://doi.org/10.1016/j.jcc.2019.02.004

Doe, J. (2021, April 5). The future of AI in cybersecurity. Tech Insights. https://www.techinsights.com/future-of-ai-cybersecurity

Lee, H. J. (2022, June). Enhancing machine learning algorithms for data security. In R. T. Nelson (Ed.), Proceedings of the International Conference on Data Security (pp. 45-60). Data Science Press. https://doi.org/10.1016/j.icds.2022.06.003

Smith, J. A. (2020). Understanding encryption: Methods and applications. Cyber Press.

Tips for Creating an APA References Section

- Double-Check Details
  - Ensure that all author names, publication years, titles, and sources are correct.
- Use Hanging Indentation
  - Format each reference with a hanging indent, where the first line is flush left, and subsequent lines are indented.
- Include DOIs and URLs
  - Provide DOIs for journal articles and URLs for online sources when available.
- Follow Capitalization Rules
  - For titles of books and articles, only the first word and proper nouns are capitalized. For journal titles, capitalize all major words.
- Alphabetize by Last Name
  - Arrange the references alphabetically by the last name of the first author.

The ABSTRACT

The abstract is a brief summary of your entire research paper. It provides readers with a quick overview of the study, including the research problem, methods, results, and conclusions. The abstract is crucial as it helps readers decide whether to read the full paper.

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Importance of a Well-Written Abstract

- First Impression
  - The abstract is often the first part of your paper that readers see. A well-written abstract can capture their interest and encourage them to read further.
- Concise Summary
  - It provides a concise summary of the research, making it easier for readers to understand the purpose and significance of the study quickly.
- Indexing and Retrieval
  - Abstracts are used by indexing databases and search engines, making your research more discoverable.
- Guidance for Readers
  - It helps readers quickly determine if the paper is relevant to their interests and needs.

Key Components of an Abstract

- Background and Objective
  - Briefly introduce the research topic and state the primary objective of the study.
  - Example: "With the rapid growth of cloud computing, ensuring data security has become increasingly critical. This study aims to develop an advanced encryption algorithm to enhance data security in cloud storage systems."
- Methods
  - Summarize the research methods used to conduct the study.
  - Example: "The study followed the Software Development Life Cycle (SDLC) framework, including planning, analysis, design, implementation, and testing phases. Expert testing and user usability testing were conducted using McCall's software quality model and the USE Questionnaire."
- Results
  - Highlight the key findings of the research.
  - Example: "The new encryption algorithm significantly improved data security, achieving an average encryption strength of 256 bits. Statistical analysis indicated a significant performance improvement ($p < 0.01$)."
- Conclusion
  - State the main conclusions and implications of the study.
  - Example: "The advanced encryption algorithm provides a robust solution for enhancing data security in cloud storage systems, offering significant improvements over existing methods. Future research should explore its integration with comprehensive security frameworks."

Example Abstract for IT/IS Research

Abstract

With the rapid growth of cloud computing, ensuring data security has become increasingly critical. This study aims to develop an advanced encryption algorithm to enhance data security in cloud storage systems. The research followed the Software Development Life Cycle

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(SDLC) framework, including planning, analysis, design, implementation, and testing phases. Expert testing and user usability testing were conducted using McCall's software quality model and the USE Questionnaire. The new encryption algorithm significantly improved data security, achieving an average encryption strength of 256 bits. Statistical analysis indicated a significant performance improvement ($p < 0.01$). The advanced encryption algorithm provides a robust solution for enhancing data security in cloud storage systems, offering significant improvements over existing methods. Future research should explore its integration with comprehensive security frameworks.

Tips for Writing an Effective Abstract

- Be Concise
  - Keep the abstract between 150-250 words. Ensure it is concise yet comprehensive enough to cover all key aspects of the study.
- Use Clear and Simple Language
  - Avoid jargon and complex sentences. Use clear and straightforward language to make the abstract accessible to a broader audience.
- Focus on Key Points
  - Emphasize the most important aspects of your research, including the problem, methods, results, and conclusions.
- Avoid References
  - Do not include citations or references in the abstract. The abstract should be self-contained.
- Review and Revise
  - Review and revise the abstract multiple times to ensure it accurately reflects the content of the paper and is free of errors.

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CAPSTONE PROJECT GUIDELINES

Scope of the Capstone Project

The Capstone Project should integrate the different courses, knowledge, and competencies learned in the curriculum. Students are encouraged to produce innovative results, generate new knowledge or theories, or explore new frontiers of knowledge or application areas.

For Information Technology Capstone Projects, recommended infrastructure and its implications on other systems should be clearly specified in the final report with the introduction of the project.

For Information Systems Capstone Projects, changes in process and information flow and/or information policies with the introduction of the system should be clearly specified in the final report.

The capstone project adviser should determine the appropriate complexity level of the specific problem being addressed and the proposed solution, considering the duration of the project, the composition of the team, and the resources available.

Suggested Areas for Capstone Projects

The following is a list of some suggested areas per program per CMO 25 s2015. The specific areas identified for each degree program may also be considered for the other computing degree programs, depending on the scope, limitations, and approach and following the principles stated in preceding sections.

IT Capstone Project

- focuses on the infrastructure, application, or processes involved in introducing a computing solution to a problem.
- hence, this project **includes a comprehensive report on infrastructure requirements & the project's implications on other existing systems**.

IS Capstone Project

- focuses on business processes and the implications of introducing a computing solution to a problem.
- hence, the **changes on the affected business processes, information flow, and information policies must be discussed clearly** in the terminal report.

Information Technology

- Software Development
  - Software Customization
  - Information Systems Development for an actual client (with pilot testing)
  - Web Applications Development (with at least alpha testing on live servers)
  - Mobile Computing Systems
- Multimedia Systems
  - Game Development
  - eLearning Systems

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    - Interactive Systems
    - Information Kiosks
- Network Design and Implementation and Server Farm Configuration and Management
- IT Management
    - IT Strategic Plan for sufficiently complex enterprises
    - IT Security Analysis, Planning and Implementation

Information Systems

- Software Development
    - Software Customization
    - Information Systems Development for an actual client
    - Web Applications Development
    - Mobile Computing Systems
- IS Planning
    - Enterprise Resource Plan
    - Information Systems Strategic Plan
- Analysis and Design of a sufficiently complex business system

Capstone Project Duration

Students are given ample time to finish their capstone project. Students will enroll their Capstone 1 during their 3rd Year Summer, and their Capstone 2 on their 4th Year 1st Semester.

Composition of Capstone Project Groups

Students should preferably work in teams of two (2) to four (4) members depending on the complexity of the project. The adviser should be able to determine whether the team can complete the project on time.

Multidisciplinary teams are also encouraged, provided that team members prepare separate documentations per program.

Adviser / Panel Composition

Panel Composition

The capstone project is prepared under the guidance of an adviser and presented and accepted by a panel composed of the following:

- College Dean (Chair)
- Program Head
- College/Program Research Coordinator
- Research Director
- External Expert
- Adviser

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In cases wherein the adviser of the group is the College Dean, the Program Head will be the Chair during the presentation.

Adviser / Panel Qualifications

The adviser must have completed a computing project successfully beyond the bachelor's degree project. The adviser should be a full-time faculty member of the University.

Advisers and Panel Members should have a degree in a Computing or allied programs (except for the Research Director), or must be domain experts in the area of study. At least one of the panel members must have a master's degree in Computing (preferably in the same field as the Capstone Project) or allied program. At least one of the panel members should preferably have industry experience.

The adviser must be able to guide the students throughout the whole project life cycle, including the capstone project defense and possible project deployment.

Faculty advisers should preferably handle at most 3 capstone projects at one time, and in no case should exceed 10 projects.

In case of the participation of an external client, then the organization for which the project is intended should be represented as much as possible.

Presentation of the Capstone Projects

The Capstone Project must be presented in a public forum. This forum may be an international, national, regional or school-based conference, meeting, or seminar that is announced and open to interested parties. This may be separate from the presentation before the Panel as mentioned in the previous section. A school-based colloquium organized for this purpose would suffice to satisfy this requirement. Presentation in a public forum, such as the International Conference on IT Education (ICITE) of PSITE, is encouraged.

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THE CAPSTONE PROJECT FORMAT

Note: Template will be provided.

ABSTRACT

- Maximum of 250 words

KEYWORDS

INTRODUCTION

- 3 to 4 paragraphs
- Should include:
  - Project Context, Purpose and Description
  - Gap / Problem identified
  - Closest prior art (at least 2)
  - The significance of this study
  - Scope and Limitations

OBJECTIVES OF THE STUDY

- The general and specific objectives in paragraph form (system-based & research-based)

MATERIALS AND METHODS

- Research Design (descriptive-developmental)
- SDLC Model (Please note, this will vary depending on the model you choose)
  - Example - for Agile

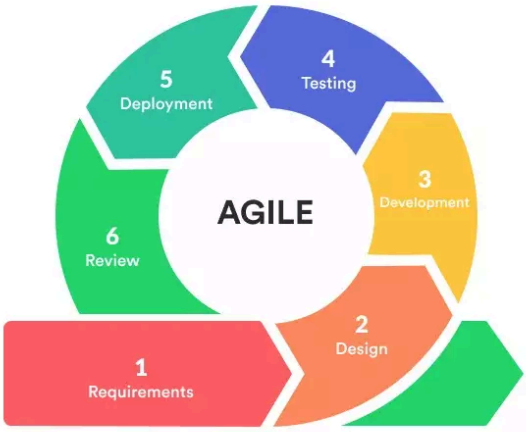

Fig. 1. Agile Software Development Life Cycle Model

  - Requirements Phase
    - Discuss your Initial data gathering process / needs assessment
    - Initial Data Gathering Instrument.
    - Data Gathering and Analysis

        - System Development and System Implementation Requirements
    - Design Phase
        - Discuss the process of designing the systems
        - Include your Use-Case Diagram/ data models and System Architecture (please note that each figure should be discussed in paragraph form)
    - Development Phase
        - Present the Programming Language, Framework (if applicable), and Libraries
        - Present the DBMS / Storage of data (if applicable)
        - Discuss the process of building the software according to design specifications
        - Discuss the integration of system components
    - Testing Phase
        - Present the instrument you used for testing (ex. McCalls, SQuaRE, USE/CSUQ/TAM)
        - Present the process of unit testing (if applicable)
        - Present the process of system testing
            - Expert testing
            - Usability testing
    - Deployment Phase
        - Present the process of finalization and implementation of the system
        - Present your deployment plan (in paragraph form)
    - Review Phase
        - Present the process of reviewing the whole process and how to address issues and implement improvements

***** THIS IS A GUIDE ONLY: Phases may be different from the model you choose*****
***** Please note that phases may be merged if your discussion may cover 2 or more SDLC phases in the model*****

- Ethical Consideration

RESULTS AND DISCUSSION

- Figures showing the system feature in the objectives followed by discussion for each figure
- Table for the result of the Quality of the System

Table 1. The quality of the system based on ISO/IEC 25010 Systems and Software Quality Requirements and Evaluation (SQuaRE)

Criteria	Mean	Interpretation
Grand Mean		

- Discussion for the Table of Quality of the System

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Table 2. The level of usability of the system based on Computer System Usability Questionnaire (CSUQ)

Criteria	Mean	Interpretation
Grand Mean		

- Discussion for the Table of System Usability

CONCLUSION AND RECOMMENDATION

- Paragraph form
- Include implications in last paragraph

ACKNOWLEDGEMENT

- This is optional. You may acknowledge people or organizations that contributed/ helped in this capstone project.

REFERENCES

- APA Style
- Hanging indent

APPENDICES

- Research Instruments
- Communication Letters
- Consent Form (if applicable)
- Timetable for the Research (Gantt Chart)
- Relevant Source Code (Note: include only code/algorithm that solves the gap you presented, not the HTML/CSS etc.)
- User's Manual
- Researchers Profile

NOTE:
As per **CMO 25 s2015**:

IT Capstone project should include a comprehensive report on infrastructure requirements and the project's implications on other existing systems.

IS Capstone Project should include discussion on the changes on the affected business processes, information flow, and information policies.

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Book Binding Guidelines

- Upon completion of the Capstone Project and its manuscript, the students shall be required to submit 2 copies of the manuscript with signed approval sheet. (Note that the developed system (output) and other components (if applicable), including the softcopy of the manuscript (in pdf/docx format) should be submitted also)
- A "Group Code" will be assigned to each capstone project group.
    - Example: BSIT 4A
        - Group 1: A1
        - Group 2: A2
    - Example: BSIT 4B
        - Group 1: B1
        - Group 2: B2
- Page Numbering of each group shall follow the following format:
    - <Group Code> - <Page Number>
    - Example instead of "1" for page 1 of group A1:
        - The page number is: A1 - 1
        - A1 - 2
        - A1 - 3 … and so on…
    - or:
        - A1 [i]
        - A1 [ii] … and so on …
- Font Size: 11
- Font: Time New Roman
- Spacing: Single (Follow the Template Provided)
- Order of Pages per group
    - Title Page
    - Approval Sheet
    - Table of Contents
    - List of Table
    - List of Figures
    - Full paper (in IMRaD Format)
    - Appendices
    - Curriculum Vitae
- Hardbound Numbering
    - In case that there are multiple books per batch:
        - Each Book will be labeled with the following:
            - Volume # | Number # | Year
            - Example:
                - Volume 1 | Number 1 | 2024
                - Volume 1 | Number 2 | 2024
                - Volume 2 | Number 1 | 2025
                - Volume 2 | Number 2 | 2025
                - and so on…
- Hardbound Contents
    - After the Hardbound Cover Page, the contents page will follow
    - The Content Page lists the different capstone project in the compilation (book)

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- In the Content Page, the Capstone Project Title will be preceded by the Group Code
  - Example:
    - A1 WEB-BASED SMART CAFE MANAGEMENT SYSTEM
      Author 1 | Author 2 | Author 3 | Author 4
    - A2 WEB-BASED CUSTOMIZED APPAREL DESIGN AND ORDERING SYSTEM
      Author 1 | Author 2 | Author 3 | Author 4
    - and so on…

- A book may contain from 6 to 15 Manuscripts depending on the thickness. The program head will decide on the number of books per batch, and the number of manuscripts per book.

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THE TITLE PAGE

{2 line spacing from top margin}

TITLE TITLE TITLE TITLE TITLE TITLE TITLE TITLE TITLE
TITLE TITLE TITLE

{8 line spacing from the title}

A Capstone Project

{8 line spacing from the line above}

presented to
The Faculty of the College of Information and Communications Technology and Engineering
State University of Northern Negros
Old Sagay Campus, Sagay City, Negros Occidental

{8 line spacing from the line above}

In Partial Fulfillment
of the Requirements for the Degree
BACHELOR OF SCIENCE IN INFORMATION _____

{8 line spacing from the line above}

FIRSTNAME M. LASTNAME
FIRSTNAME M. LASTNAME
FIRSTNAME M. LASTNAME
FIRSTNAME M. LASTNAME

Month Year
{align at the bottom of the page}

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THE BOOK COVER PAGE

{4 line spacing from top margin}

A COMPILATION
OF UNDERGRADUATE CAPSTONE PROJECTS

{8 line spacing from the line above}

Presented to
The Faculty of the College of Information and Communications Technology and Engineering
State University of Northern Negros
Old Sagay Campus, Sagay City, Negros Occidental

{8 line spacing from the line above}

In Partial Fulfillment
of the Requirements for the Degree
BACHELOR OF SCIENCE IN INFORMATION _____

{8 line spacing from the line above}

Volume __ | Number __ | Year
{align at the bottom of the page}

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THE CONTENTS PAGE

CONTENTS

A1. TITLE OF CAPSTONE PROJECT 1
Author 1 | Author 2 | Author 3 | Author 4

A2. TITLE OF CAPSTONE PROJECT 2
Author 1 | Author 2 | Author 3 | Author 4

A3. TITLE OF CAPSTONE PROJECT 3
Author 1 | Author 2 | Author 3 | Author 4

A4. TITLE OF CAPSTONE PROJECT 4
Author 1 | Author 2 | Author 3 | Author 4

A5. TITLE OF CAPSTONE PROJECT 5
Author 1 | Author 2 | Author 3 | Author 4

A6. TITLE OF CAPSTONE PROJECT 6
Author 1 | Author 2 | Author 3 | Author 4

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BOOK SIDE PRINT

COURSE	A COMPILATION OF UNDERGRADUATE CAPSTONE PROJECTS	VOLUME _ | NUMBER _ | 202_

EXAMPLE

BSIT	A COMPILATION OF UNDERGRADUATE CAPSTONE PROJECTS	VOLUME 1 | NUMBER 1 | 2024

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References

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Glasman-Deal, H. (2010). Science research writing for non-native speakers of English: A guide for non-native speakers of English. Imperial College Press.

Nair, R. B., & Nair, V. D. (2014). Scientific writing and communication in agriculture and natural resources. Springer. https://doi.org/10.1007/978-3-319-03103-0

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Swales, J. M., & Feak, C. B. (2012). Academic writing for graduate students: Essential tasks and skills (3rd ed.). University of Michigan Press.

Zinsser, W. (2006). On writing well: The classic guide to writing nonfiction (30th Anniversary ed.). Harper Perennial.

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